Image Classification with Transfer Learning

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Introduction/Background Check

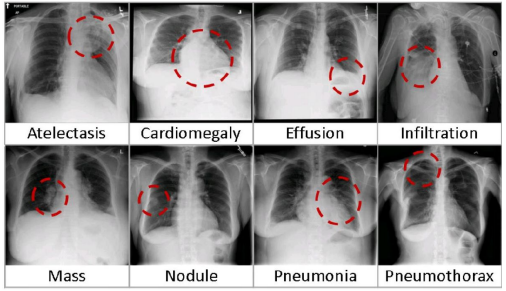


Image Classfication

The task of assigning labels to an image based on its content

Applications:

- Medical Imaging
- Autonomous Driving
- Security Systems

Challenges:

- Requires large labeled datasets
- CNNs are computationally expensive

What's different from our approach?

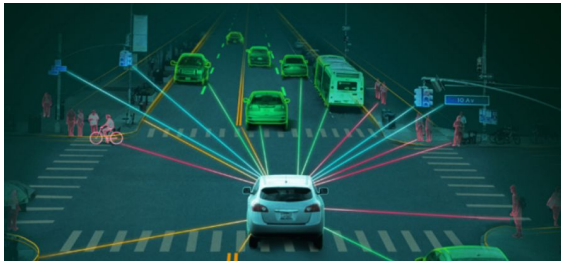
Answer: Transfer Learning

Transfer Learning Approach

- Leverages pretrained CNN models on large datasets (ImageNet)
- Reuse learned feature representation for new tasks
- Reduces training time and data requirements
- Avoids training deep networks from scratch

Applied Through:

- Feature Extraction
- Fine Tuning



Overview of Transfer Learning

Pretrained Models

Models trained on large datasets (e.g., ImageNet with 14M+ images) capture ruch, generalizable feature representations

Feature Extraction

Freeze all pretrained convolutional layers. Only final classification head is trained on the new task

Fine-Tuning

Unfreezes some of all upper layers. Retrain with a small learning rate to adapt features to target domain.

Why It Works

Low-level features (edges, textures) transfer universally. High-level features are domain specific and benefit most from fine-tuning

Related Work

Pretrained backbones: ResNet and VGG16 have been widely adopted for image classification

Data Efficiency: Transfer learning enables adaptation to new datasets with limited data

Performance Gains: Prior research shows improved accuracy and reduced training time VS training from scratch

2 Strategies: Fine-tuning and feature extraction are most common adaptation approaches

Domain Similarity: Performance is strongly influenced by how closely the source and target datasets match.

Related Work & References

ResNet – He et al. (2016)

Introduced residual (skip) connections to solve the vanishing gradient problem, enabling networks of 50-152 layers to train effectively. Established as the go-to baseline for transfer learning tasks

VGG – Simonyan & Zisserman (2015)

Demonstrated that network depth with small 3 x 3 convolution filters is a key driver of performance. Paved the way for the deep architecture paradigm used in modern CNNs.

EfficientNet – Tan & Le (2019)

Proposed compound scaling – systematically scaling depth, width, and resolution together. Achieves strong accuracy at significantly lower computational cost, making it ideal for resource-constrained experiements

Project Motivation

High Cost of Training: Deep CNNs require massive labeled datasets and expensive compute

Limitations of Current Transfer Learning: Most models are evaluated on datasets similar to their pre training domain

Domain Shift Challenge:

- Real-world data often differs significantly from ImageNet-like distributions
- Domain generalization is critical for real deployment but is rarely tested in controlled academic settings

Performance Degradation: Models can fail or lose accuracy under new conditions

Why it Matters: Understanding these tradeoffs has direct implications for a low-resource applications like medical imaging and robotics

Open Question: The effectiveness of different fine-tuning strategies under domain shift is still not well understood

Project Goals

Compare feature extraction vs fine-tuning to assess the impact on performance and efficiency

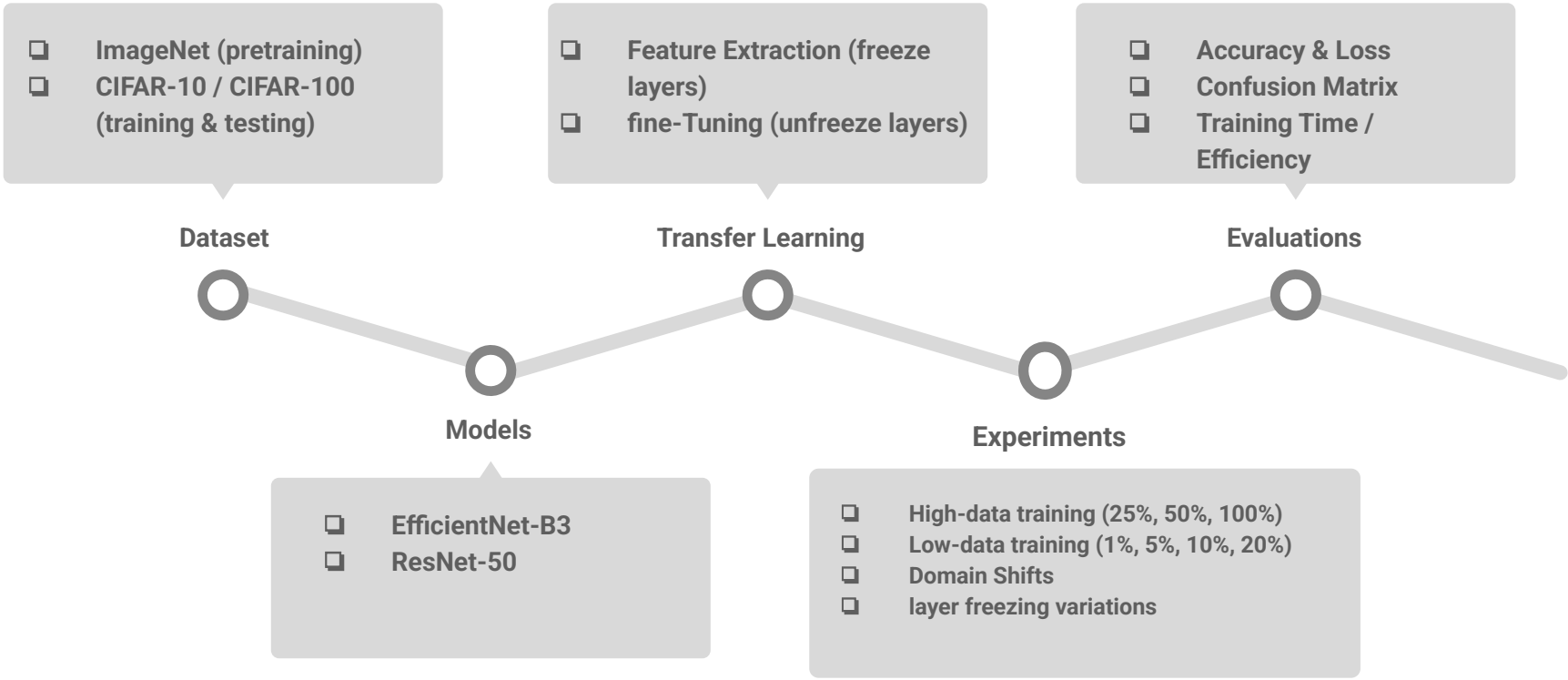
Leverage pretrained ResNet-50 and EfficientNet-B3 for image classification on CIFAR-10 and CIFAR-100, training on 25%, 50%, and 100% data splits to simulate low-data scenarios

Evaluate cross-dataset generalization by training on CIFAR-10 and testing on CIFAR-100 to measure domain shift impact

Systematically compare five freezing levels (0–100%) to identify the optimal fine-tuning strategy per architecture

Evaluate accuracy, loss trends, and training efficiency across models and conditions for practical insights

Methodology Overview



Dataset

ImageNet(Subset)
Pre Training Source

- 14M+ images total
- 1000 object categories
- High resolution, high variability
- Used as pretrained weights – NOT retrained

CIFAR-10
Primary Training Dataset

- 60,000 images
- 10 classes (airplane, dog, ship, etc.)
- 32 x 32 pixel resolution
- 50k train / 10k test split

CIFAR-100
Domain Shift Target

- 60,000 images
- 100 fine-grained classes
- 32 x 32 pixel resolution
- Same size as CIFAR-10, harder task

Models

ResNet-50 (baseline)

- 50-layer deep residual network
- Residual/skip connections prevent vanishing gradients
- ~25M parameters
- Strong, well-understood baseline for comparison

EfficientNet-B3 (secondary)

- Compound scaling: depth + width + resolution
- ~12M parameters – more efficient than ResNet-50.
- Higher accuracy per FLOP on ImageNet benchmarks
- Tests whether backbone choice affects results under same conditions

	ResNet-50	EfficientNet-B3
Params	~25M	~12M
ImageNet Top-1	76.1%	81.6%
Depth	50 layers	26 layers

Transfer Learning

Fine-Tuning

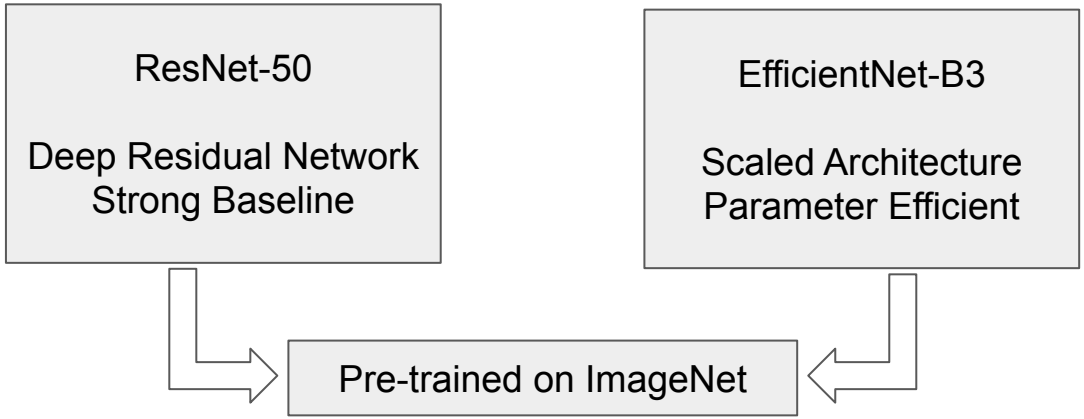
- Pre-trained weights used in initialization
- Selected layers unfrozen and trained
- Allows model to adapt to new dataset
- Small learning rate used to avoid overwriting pretrained knowledge

Feature Extraction

- Pre-trained model used as fixed feature extractor
- All convolutional layers frozen
- Only final classification layer trained
- Faster training, fewer parameters updated

Low-level features (edges, textures) transfer universally – high level features are domain specific and benefit most from fine-tuning

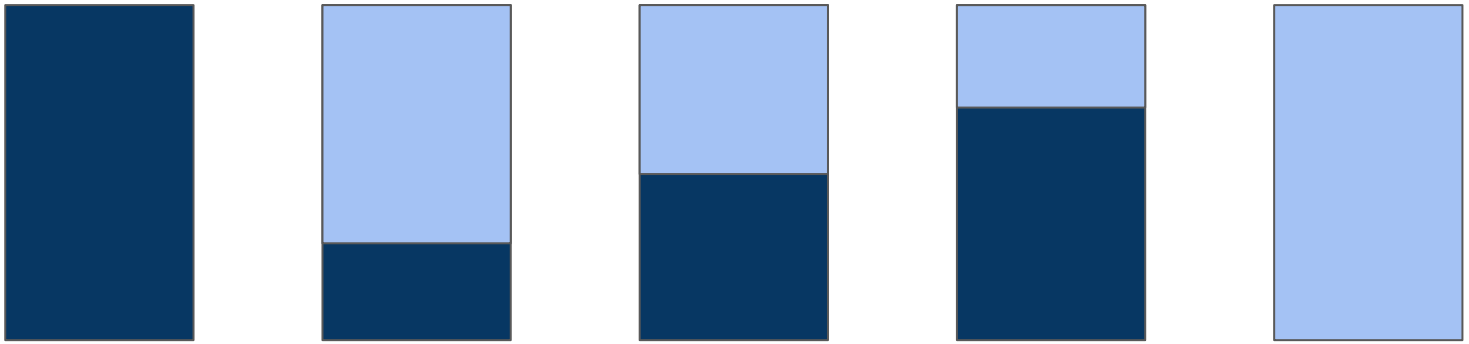
Architectures



Hyperparameter Tuning

- Optimizer: AdamW
- Learning Rate: 0.001
- BatchSize: 64
- Epochs: 15
- Loss Function: Cross Entropy

Experiment 1: Layer Freezing Coverage



0%

Full
Fine-Tune

25%

Mostly
Unfrozen

50%

Half
Frozen

75%

Mostly
Frozen

100%

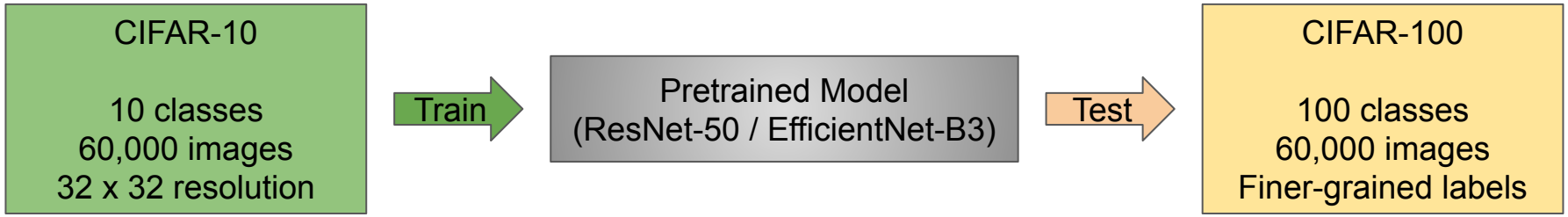
Feature
Extract

Measuring: Accuracy + training time per level

Hypothesis: Intermediate freezing levels (25-50%) will outperform both extremes. General low level features are preserved but upper layers adapt to CIFAR

Testing: Impact of freezing depth on model performance

Experiment 2: Domain Shift



Cross-Dataset Accuracy

How accurately does a CIFAR-10 trained model classify CIFAR-100 images with no retraining

Performance Degradation

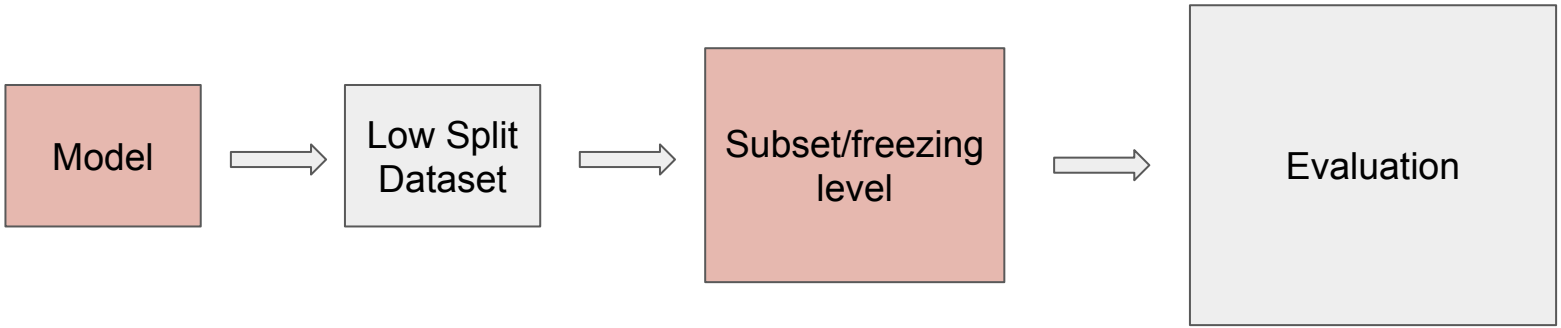
How much does accuracy drop vs a model trained directly on CIFAR-100

Architecture Sensitivity

Does ResNet-50 or EfficientNet-B3 generalize better under domain shift

Hypothesis: Feature extraction will generalize better than full fine-tuning under domain shift

Experiment 3: (Data Size vs Layer Freezing)



Measuring: Accuracy, loss, performance, size impact with freezing

Hypothesis: Partial tuning with transfer learning will generalize better than full fine-tuning and feature extraction under a lower data split size

Metric Evaluation

Accuracy

- ❏ Top-1 validation accuracy per condition
- ❏ Compared across models, freeze levels, and data splits
- ❏ Primary metric for all 60 runs

Loss Curves

- ❏ Training vs validation loss per epoch
- ❏ Shows overfitting behavior – does val loss diverge from train loss
- ❏ Especially important for low data splits (25%)

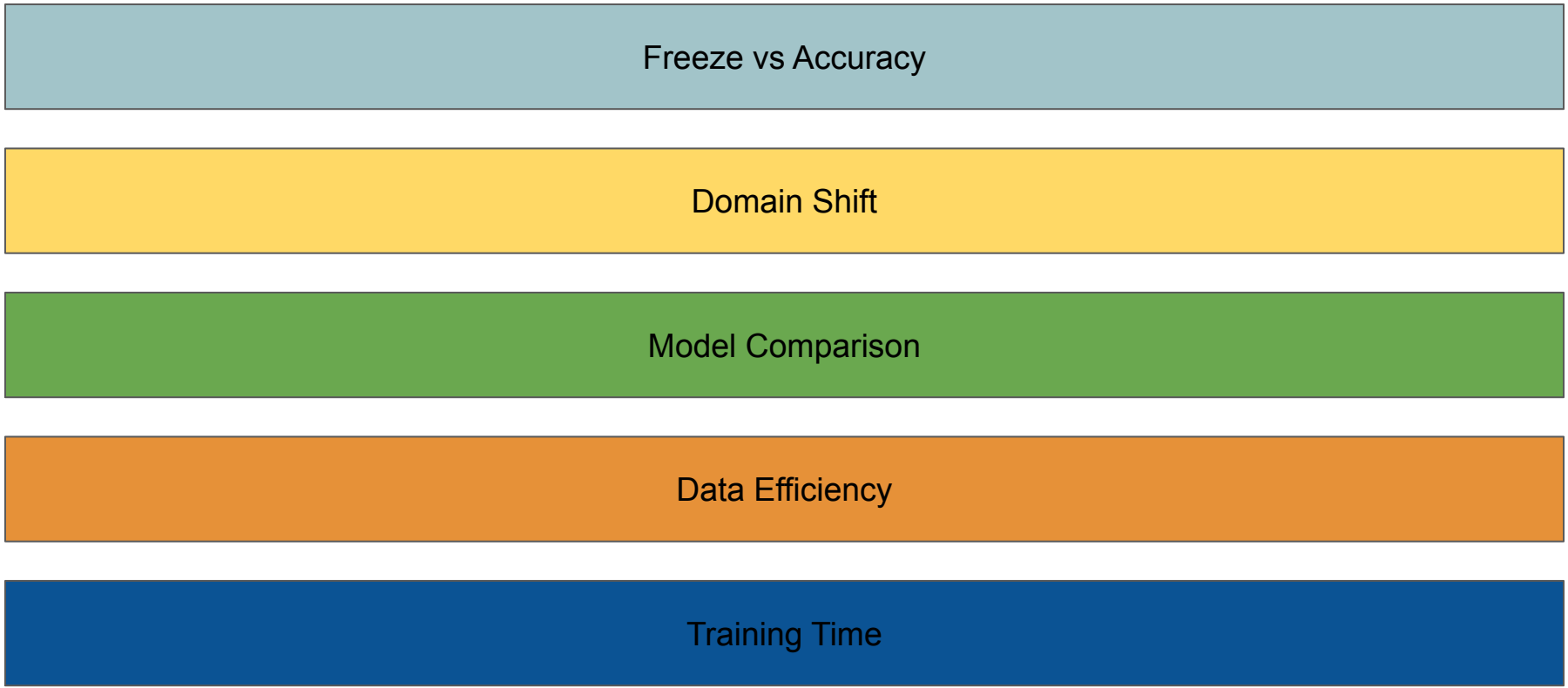
Training Time / Efficiency

- ❏ Wall-clock time per run
- ❏ Trainable parameter count per freeze level
- ❏ Shows cost vs accuracy tradeoff between models

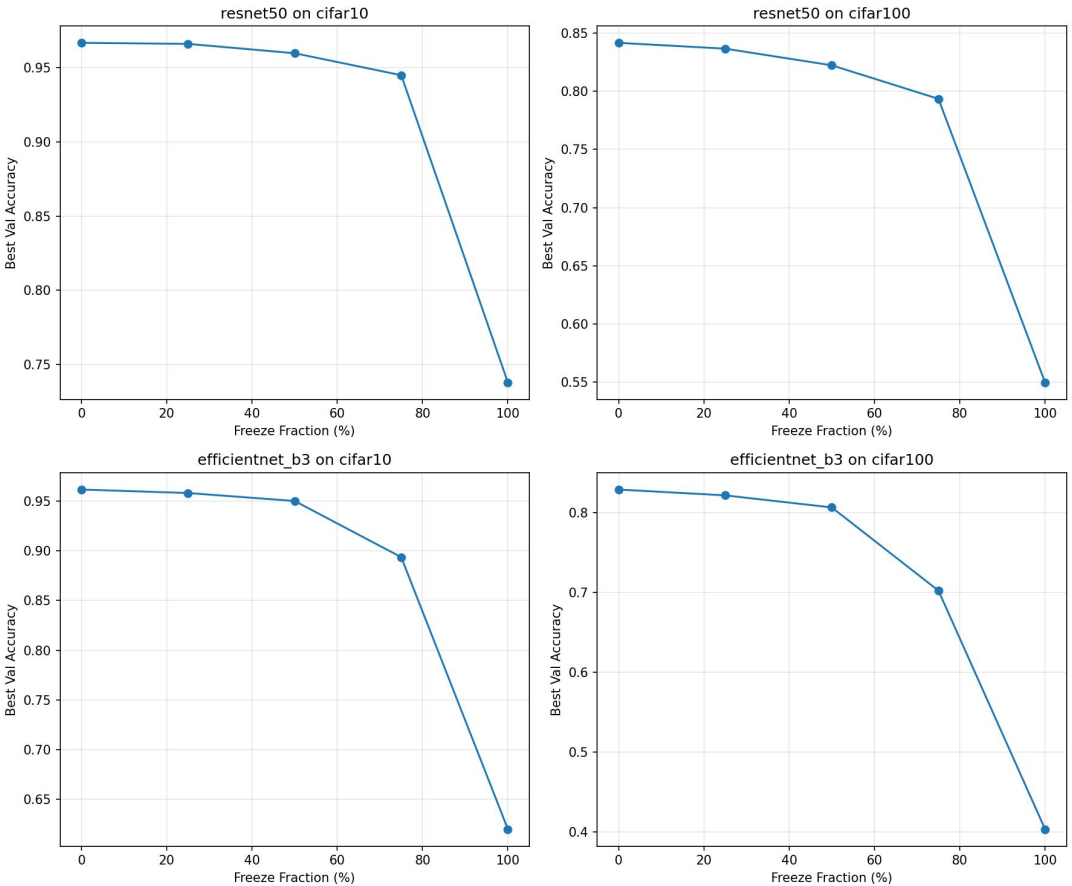
Domain Shift accuracy

- ❏ CIFAR-10 trained → CIFAR-100 superclass accuracy
- ❏ Which freeze strategy generalizes best across datasets

Visuals for Results (High-Data)



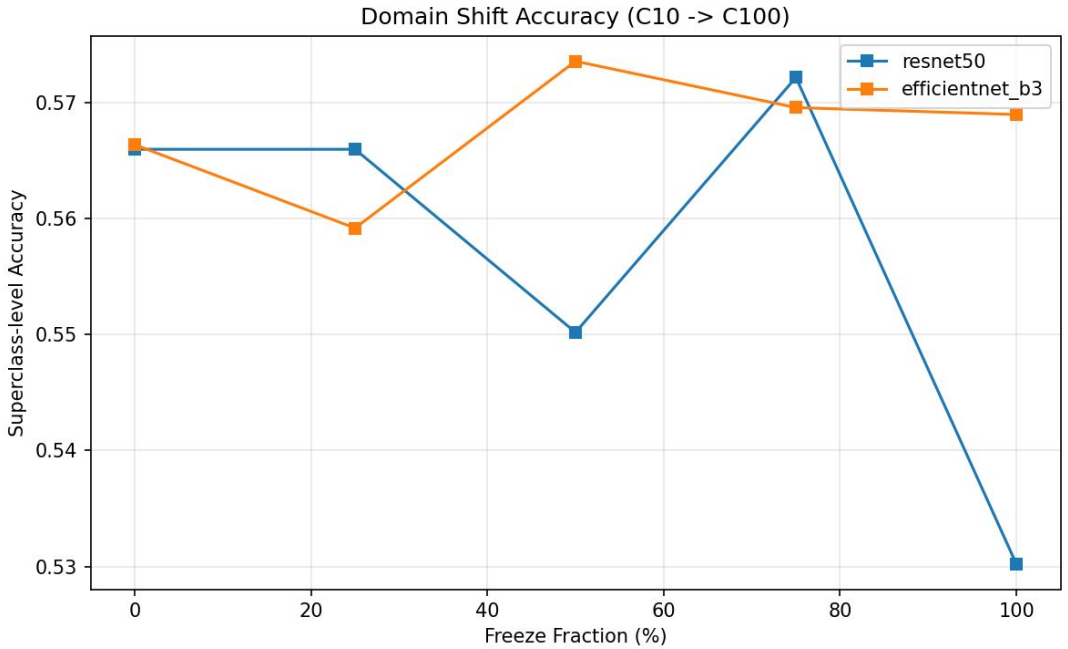
Freeze vs Accuracy



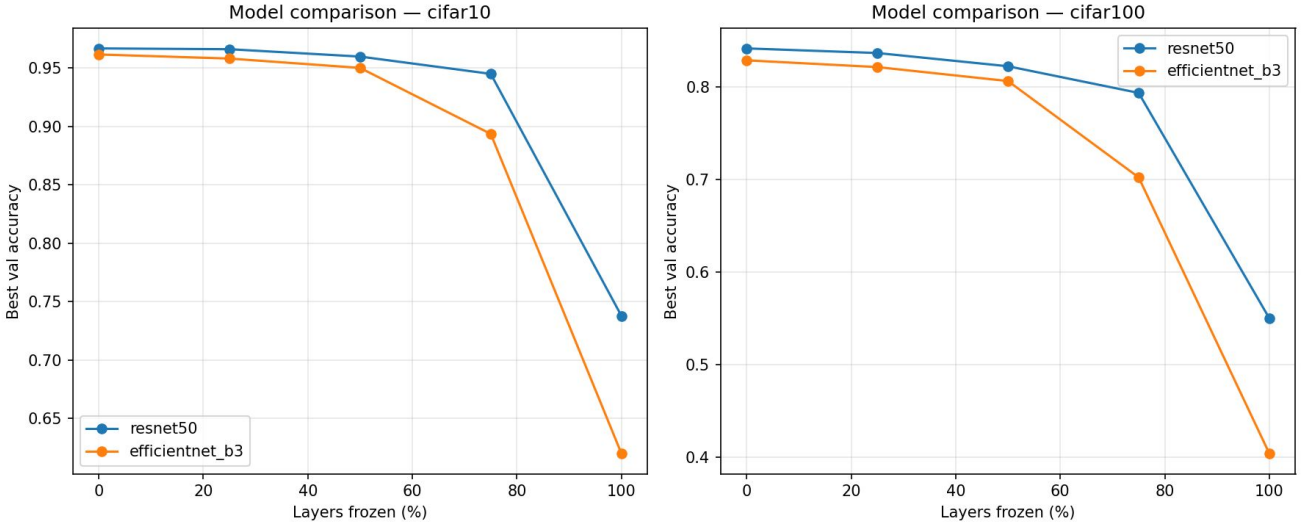
- Full fine-tuning (0% frozen) consistently achieves highest accuracy across all conditions
- Accuracy drops sharply after 75% freezing – feature extraction (100%) performs worst
- ResNet-50 on CIFAR-10 reaches 96.6% with full fine-tuning vs 73.7% with feature extraction

Domain Shift

- Both models start similarly — at 0% frozen both ResNet-50 and EfficientNet-B3 achieve ~56.6% superclass accuracy, showing comparable baseline generalization.
- Interesting crossovers: EfficientNet-B3 temporarily drops below ResNet-50 at 25% freeze before recovering, and both models converge at 75% freeze before diverging sharply, with ResNet-50 collapsing to 53.02% at full feature extraction while EfficientNet-B3 maintains stable generalization at 56.90%.
- EfficientNet generalizes better under full feature extraction
- This suggests EfficientNet's compact features transfer more broadly than ResNet's



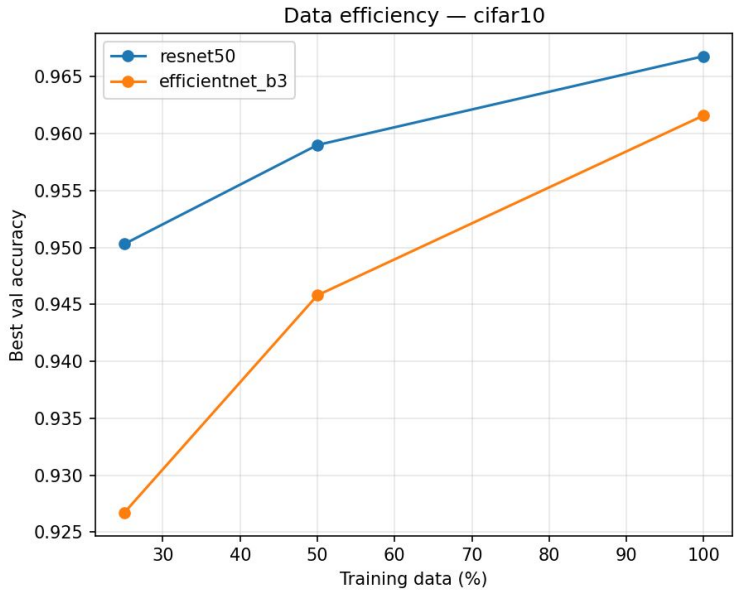
Model Comparison

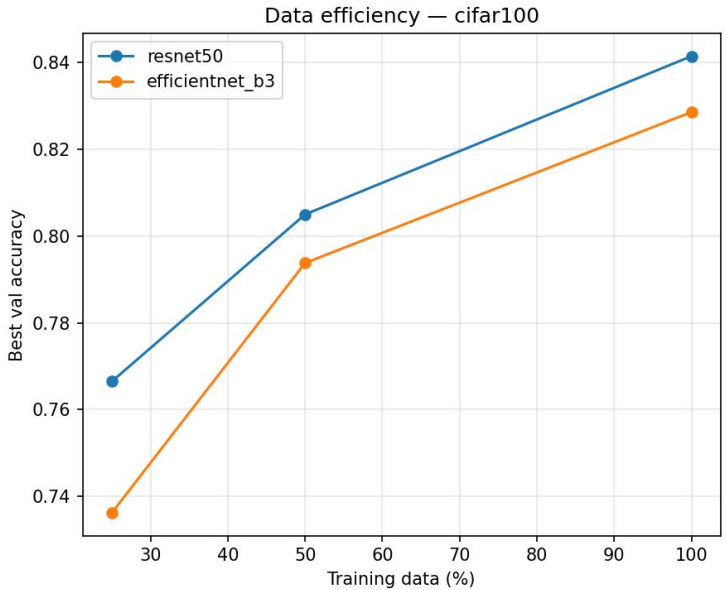


- ResNet-50 outperforms EfficientNet-B3 across all freeze levels on both datasets
- Gap widens significantly at higher freeze levels – ResNet-50 is more robust to freezing
- EfficientNet drops to 59% at 100% frozen vs ResNets 74% on CIFAR-10

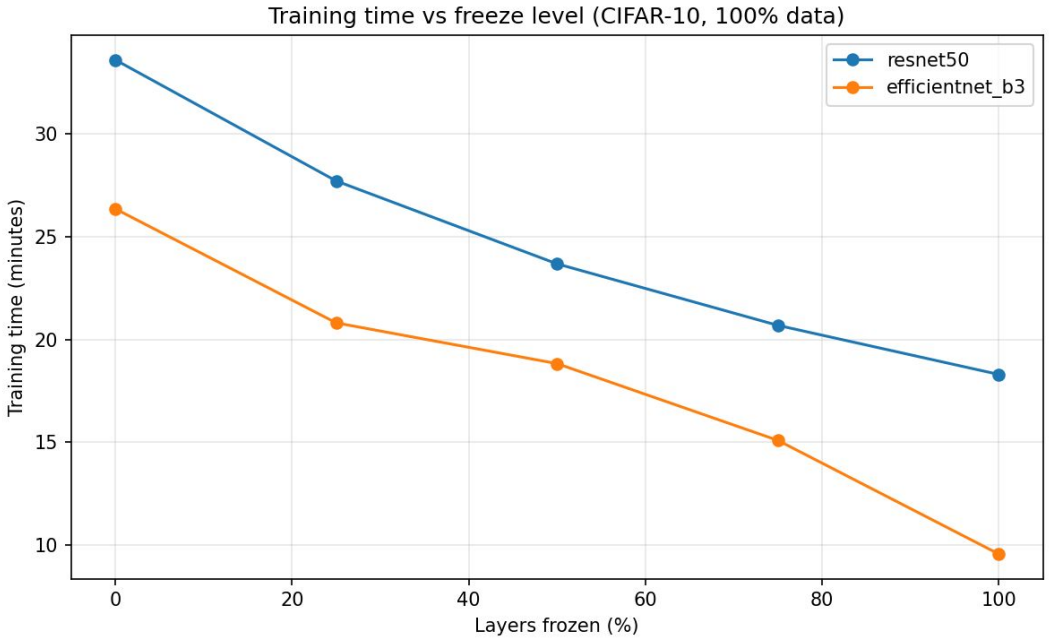
Data Efficiency

- ResNet-50 maintains 95% accuracy even with only 25% of training data on CIFAR-10
- EfficientNet is more sensitive to data size – drops to 92.75% at 25% data
- Both models improve with more data but ResNet-50 needs less data to perform well
- Demonstrates transfer learning's strength in low-data scenarios



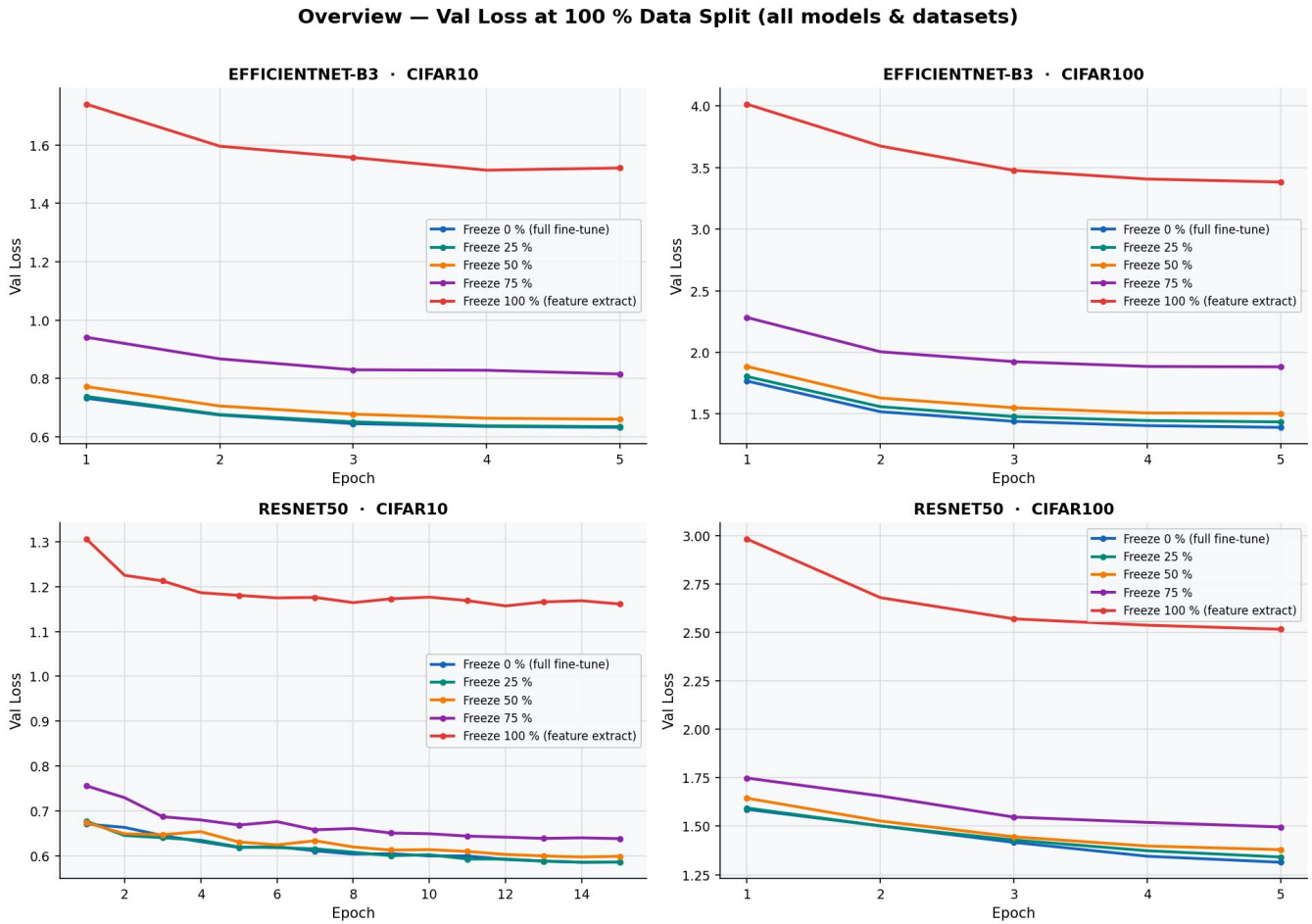


Training Time



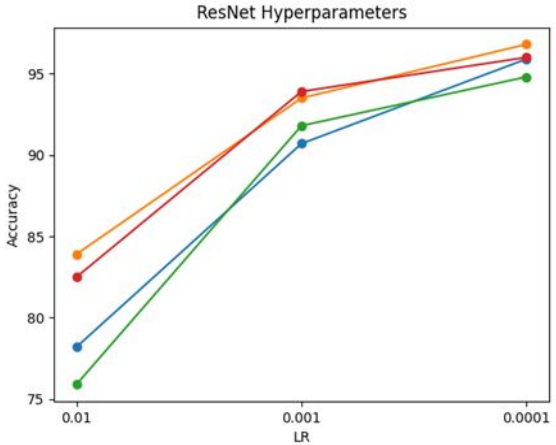
- ResNet-50 takes around 33 minutes at 0% freeze vs EfficientNet at only 26 minutes
- Both drops steadily as freeze % increase
- EfficientNet is consistently faster across all freeze levels
- Takeaway: EfficientNet is significantly more efficient even though ResNet-50 is more accurate

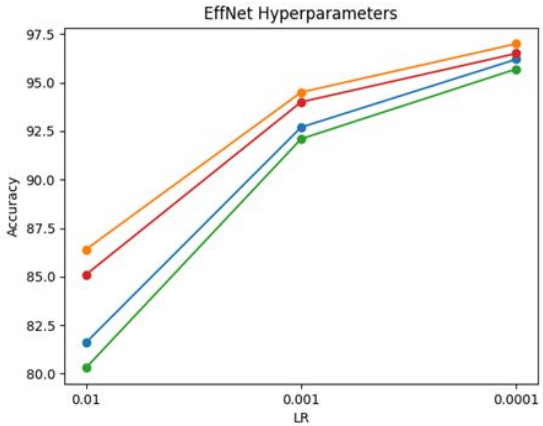
Bonus: Loss Curves



- Loss decreases steadily across epochs
- Fine-tuning achieves lowest loss
- Partial freezing shows similar convergence
- Feature Extraction (100% freeze has highest loss)
- CIFAR-100 shows higher loss than CIFAR-10

Hyper Parameter Tuning

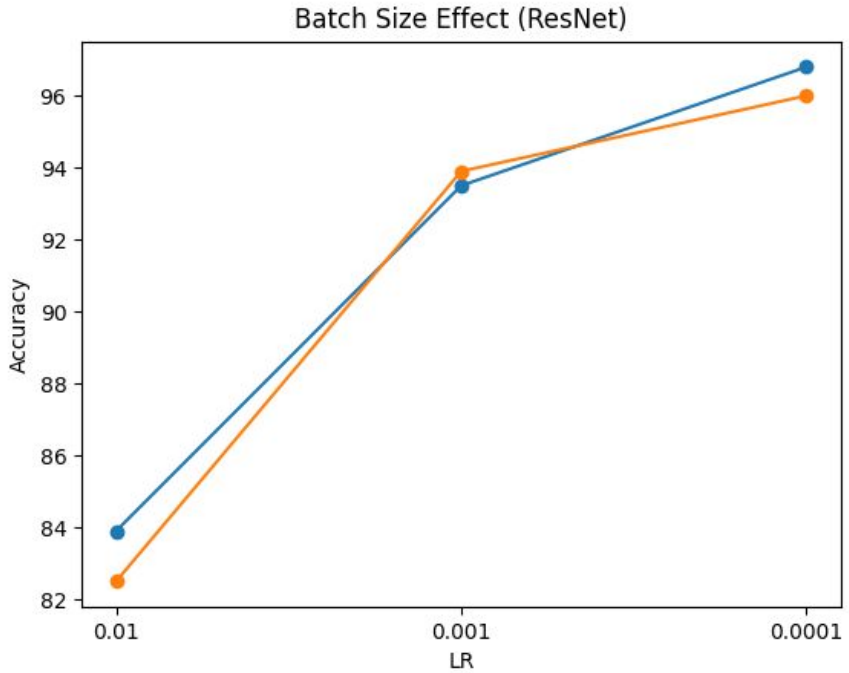




Legend:
Blue Line → Batch Size = 32, Epochs = 5
Orange Line → Batch Size = 32, Epochs = 10
Green Line → Batch Size = 64, Epochs = 5
Red Line → Batch Size = 64, Epochs = 10)

With hyper parameter tuning on ResNet accuracy improves as learning rate decreases, with smaller batch sizes and more epochs yielding better performance, and for EfficientNet it follows similar trends, achieving highest accuracy at low learning rates and outperforming ResNet across configurations.

Hyper Parameter Tuning Cont.



This graph compares the effect of batch size on model performance across different learning rates for ResNet. The results show that a smaller batch size (32) consistently outperforms a larger batch size (64), particularly at lower learning rates. This suggests that smaller batches introduce beneficial gradient noise, improving generalization. On the other hand, larger batch sizes result in slightly lower accuracy, likely due to reduced variability in gradient updates and convergence.

Results for Low Data Split

(ResNet50, CIFAR-10

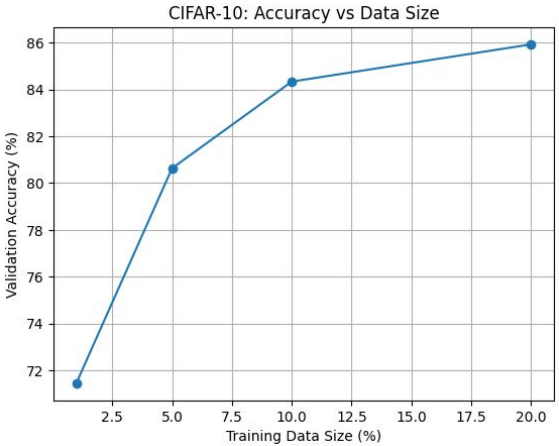
- 1% data → ~69.2% accuracy (75% frozen)
- 5% data → ~80.6% accuracy (50% frozen)
- 10% data → ~83.7% accuracy (50% frozen)
- 20% data → ~85.9% accuracy (75% frozen)

(ResNet50, CIFAR-100)

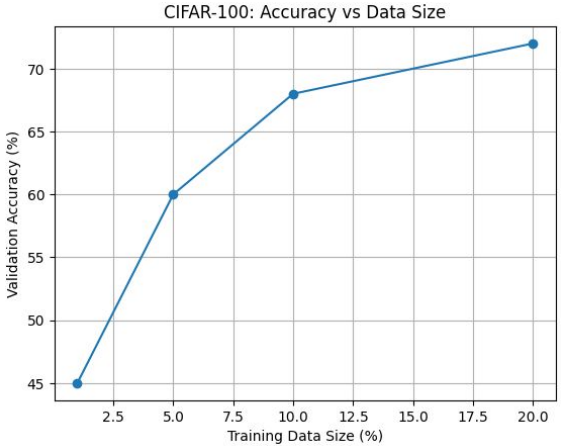
- 1% data → ~ 45% accuracy (75% frozen)
- 5% data → ~ 63% accuracy (50% frozen)
- 10% data → 72% accuracy ~(50% frozen)
- 20% data → 72% accuracy ~ (75% frozen)

- Transfer Learning is Effective (Even with Low Data)
- Accuracy increases with more data.
- Best performance consistently with 50%–75% layers frozen
- Partial fine-tuning provides the best balance between accuracy and efficiency, while full fine-tuning proves to be unstable in low data environments

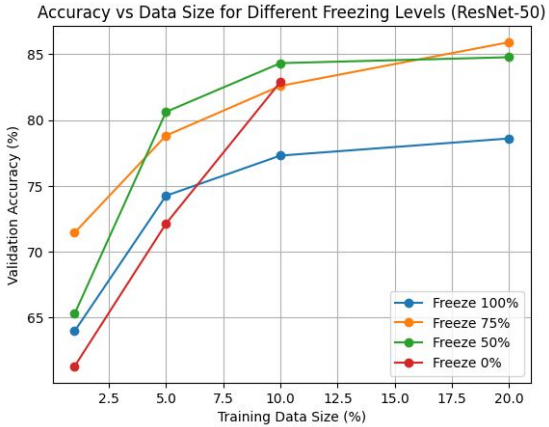
Accuracy vs Data Size (Low Split)



- Even with 1% of data the model starts with over 70% accuracy
- Accuracy improves steadily as more training data is used.



- CIFAR-100 performs lower due to higher complexity (100 classes vs 10)
- Trend is consistent and scales well with more data



- 50%–75% freezing consistently performs best
- 100% freezing performs worst overall
- 0 % freezing is unstable with low data but performs better with more data

Discussions

Fine-Tuning Depth is the Dominant Factor

- Accuracy drops sharply past 75% freezing across both models — partial fine-tuning (25–50% frozen) consistently matches full fine-tuning while using fewer resources.

Architecture Choice Depends on the Goal

- ResNet-50 achieves higher accuracy and is more robust to freezing. EfficientNet is 3–4x faster and generalizes better under domain shift — neither is strictly superior

Domain shift result was most interesting

- EfficientNet peaks at 100% frozen on CIFAR-100 while ResNet peaks at 75% — suggests EfficientNet's compact features transfer more broadly across distributions

Low-data performance validates transfer learning’s core promise

- ResNet-50 achieves ~95% accuracy on CIFAR-10 using only 25% of training data, validating the core benefit of pretrained representations

Limitations

CIFAR-10/100 are low-resolution (32×32) and may not reflect real-world complexity or high-resolution tasks

CIFAR-10 and CIFAR-100 share the same image pool — not a true domain shift; a stronger test would use completely different domains (e.g., medical or satellite imagery)

Hyperparameters (LR, batch size, epochs) were held fixed across all conditions — not optimal for every freeze level, especially feature extraction

Single runs per condition — no variance across seeds, so small accuracy differences may not be statistically meaningful

EfficientNet's domain shift advantage is not fully isolated — unclear whether it stems from architecture design, parameter count, or compound scaling

Conclusion

Partial fine-tuning (50–75% frozen) offers the best accuracy-efficiency tradeoff, especially under limited data

Full fine-tuning achieves peak accuracy but is unstable with small datasets

ResNet-50 is the stronger general-purpose backbone; EfficientNet is better suited for speed-sensitive or cross-domain applications

Transfer learning achieves near-full performance with as little as 25% of training data

Domain shift significantly degrades accuracy, but feature extraction preserves more generalizable representations than fine-tuning

Findings offer practical guidance for model selection in low-resource settings such as medical imaging and robotics

Thank You!

Questions & Answers?